

2	(a) (i) 1. 0.1 s, 0.3 s, 0.5 s, etc (<i>any two</i>)	A1 [1]
	2. either 0, 0.4 s, 0.8 s, 1.2 s or 0.2 s, 0.6 s, 1.0 s (<i>any two</i>)	A1 [1]
	(ii) period = 0.4 s frequency = $(1/0.4 =) 2.5 \text{ Hz}$	C1 A1 [2]
	(iii) phase difference = 90° or $\frac{1}{2} \pi \text{ rad}$	B1 [1]
	(b) frequency = $2.4 - 2.5 \text{ Hz}$	B1 [1]
	(c) e.g. attach sheet of card to trolley increases damping / frictional force e.g. reduce oscillator amplitude reduces power/energy input to system	M1 A1 (M1) (A1) [2]

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2 (a) (i) (tangent to line gives) direction of force on a (small test) mass R1 [1]